

OUTPUT (THASK 3)

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[44] import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

df = pd.read_csv("/content/car_data.csv")

df = df.dropna()
df = pd.get_dummies(df, columns=['fuel_type', 'selling_type', 'transmission'], drop_first=True)
X = df.drop(['car_name', 'selling_price'], axis=1)
y = df['selling_price']

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

LinearRegression
LinearRegression()

[46] predictions = model.predict(X_test)
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

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[46] predictions = model.predict(X_test)
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

print("MAE:", mean_absolute_error(y_test, predictions))
print("MSE:", mean_squared_error(y_test, predictions))
print("R2 Score:", r2_score(y_test, predictions))

MAE: 1.2163740193336217
MSE: 3.4813498305146187
R2 Score: 0.8488707839191938

[47] result = pd.DataFrame({
    "Actual Price": y_test,
    "Predicted Price": predictions
})

print(result.head(10))

Actual Price Predicted Price
177 0.35 2.954337
289 10.11 8.177163
228 4.95 6.456123
198 0.15 -1.423372
60 6.95 9.080647
9 7.45 7.417936
118 1.10 1.335139
154 0.50 0.840323
164 0.45 1.301202
33 6.00 7.490678
```



